

Experiment - 8

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Problem Statement

Given an array `arr` and an integer `k`, divide the array into contiguous subarrays such that each subarray has length $\leq k$.

Replace all elements of each subarray with the maximum element of that subarray and return the maximum possible sum.

Input

Array: `arr = [1, 15, 7, 9, 2, 5, 10]`

Integer: `k = 3`

Output

Maximum Sum = 84

Approach

Use Dynamic Programming.

Create a DP array `dp[i]` where `dp[i]` stores the maximum sum from index `i` to end.

Traverse the array from right to left.

For each index, try all subarrays of size up to `k`.

Keep track of the maximum element in the current subarray.

Update the answer using:

$\text{max_element} \times \text{length} + \text{dp}[\text{next index}]$

Store the maximum value in $dp[i]$.

Pseudo Code

1. Let n = size of array

2. Create dp array of size $n+1$ and initialize with 0

3. For i from $n-1$ to 0:

$maxVal = 0$

$best = 0$

For j from i to $\min(i+k-1, n-1)$:

$maxVal = \max(maxVal, arr[j])$

$length = j - i + 1$

$sum = maxVal * length + dp[j+1]$

$best = \max(best, sum)$

$dp[i] = best$

4. Return $dp[0]$

Time Complexity

$O(n \times k)$

Space Complexity

$O(n)$